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RESEARCH INTERESTS

Convex Analysis; Convex Optimization; Dynamical Systems; Monotone Operators

APPOINTMENTS

since 09/2025	Professor of Automatic Control and Engineering Mathematics Department of Mechanical Engineering Technische Hochschule Ingolstadt, Germany
08/2023–08/2025	Research Scientist Foundational Technologies (Autonomous Systems and Control) Siemens Technology Center Garching Siemens AG, Munich, Germany
11/2021–07/2023	Systems and Control Engineer Research and Development (System Design) MTU Aero Engines AG, Munich, Germany
09/2015–10/2021	Research and Teaching Assistant Institute for Systems Theory and Automatic Control University of Stuttgart, Germany
10/2014–07/2015	Graduate Research Assistant Department of Mechanical and Aerospace Engineering University of California, San Diego, CA, USA
06/2012–09/2012	Undergraduate Research Assistant Department of Electrical and Computer Engineering National University of Singapore, Singapore

EDUCATION

09/2015–12/2022	Ph.D., Systems Theory and Automatic Control University of Stuttgart, Germany
10/2012–08/2015	M.Sc., Engineering Cybernetics University of Stuttgart, Germany
10/2008–09/2012	B.Eng., Mechanical Engineering Deggendorf Institute of Technology, Germany

HONORS AND AWARDS

2014–2015	Dr. Jürgen und Irmgard Ulderup Fellowship
2011–2015	Fellow of German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes)
2010–2012	Fellow of German Academic Exchange Service (Deutscher Akademischer Austauschdienst)

TEACHING ACTIVITIES

Undergraduate Courses

- since 10/2025 **Engineering Mathematics I and II**, Technische Hochschule Ingolstadt
 since 10/2025 **Linear Systems Theory**, Technische Hochschule Ingolstadt
 2020–2021 **Systems and Control**, University of Stuttgart
 Winter 2017 **Introduction to Automatic Control**, University of Stuttgart

Graduate Courses

- 2016–2020 **Nonlinear Control**, University of Stuttgart
 Summer 2019 **Convex Optimization**, University of Stuttgart
 Winter 2019 **Advanced Concepts in Control**, University of Stuttgart
 Winter 2015 **Introduction to Adaptive Control**, University of Stuttgart

INVITED TALKS

- 01/2023 **Learning and Dynamical Systems Group**, Max Planck Institute for
Intelligent Systems, Tübingen, Germany
 11/2022 **Research Seminar Dynamical Systems**, Chair of Dynamical Systems,
University of Passau, Germany

PROFESSIONAL SERVICE

Membership in Professional Societies

- since 2015 **Society of Industrial and Applied Mathematics (SIAM)**
 since 2014 **Institute of Electrical and Electronics Engineers (IEEE)**
 since 2014 **Control Systems Society (CSS)**

Reviewer (Journal Articles)

Automatica; IEEE Control Systems Letters; IEEE Transactions on Automatic Control; Operations Research Letters; Optimization; SIAM Journal on Control and Optimization

PUBLICATIONS

Journal Articles

- [J-04] F. Battahi, Z. Chbani, **S. K. Niederländer** and H. Riahi, *Asymptotic behavior of the Arrow–Hurwicz differential system with Tikhonov regularization*, J. Convex Anal., 33 (2026), pp. 565–596.
 [J-03] **S. K. Niederländer**, *On the Arrow–Hurwicz differential system for linearly constrained convex minimization*, Optimization, 73 (2023), pp. 2313–2345.
 [J-02] **S. K. Niederländer**, *Second-order dynamics with Hessian-driven damping for linearly constrained convex minimization*, SIAM J. Control Optim., 59 (2021), pp. 3708–3736.
 [J-01] J. Cortés and **S. K. Niederländer**, *Distributed coordination for nonsmooth convex optimization via saddle-point dynamics*, J. Nonlinear Sci., 29 (2019), pp. 1247–1272.

Conference Proceedings

- [C-05] **S. K. Niederländer**, *The continuous steepest descent method with convex-like potential*, in Proc. IFAC World Congress, Busan, Republic of Korea, 2026, to appear.
- [C-04] **S. K. Niederländer**, *Asymptotic behavior of the nonautonomous Arrow–Hurwicz differential system*, in Proc. IEEE Conf. Decis. Control, Milan, Italy, 2024, pp. 4752-4757.
- [C-03] **S. K. Niederländer**, *Ergodic convergence results for the Arrow–Hurwicz differential system*, in Proc. IEEE Conf. Decis. Control, Singapore, 2023, pp. 7293-7298.
- [C-02] **S. K. Niederländer**, F. Allgöwer and J. Cortés, *Exponentially fast distributed coordination for nonsmooth convex optimization*, in Proc. IEEE Conf. Decis. Control, Las Vegas, NV, USA, 2016, pp. 1036-1041.
- [C-01] **S. K. Niederländer** and J. Cortés, *Distributed coordination for separable convex optimization with coupling constraints*, in Proc. IEEE Conf. Decis. Control, Osaka, Japan, 2015, pp. 694-699.

Other Works

- [O-02] **S. K. Niederländer**, *Dynamical approaches to linearly constrained convex minimization*, Ph.D. Thesis, University of Stuttgart, 2022.
- [O-01] **S. K. Niederländer**, *Distributed continuous-time coordination for nonsmooth convex and robust optimization*, Master Thesis, University of Stuttgart, 2015.

updated on July 22, 2026